FROBENIUS-PERIODIC VECTOR BUNDLES GLUE ALONG THE ALMOST BLOWUP SQUARE OF A SINGLE EQUATION, GIVING AN EQUIVALENCE OF CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EXC (sources/stacks-flat/pinned-original-source.tex line 12235).

Allowed standard prerequisites (editorial).

- The almost blowup square associated with an equation in one generator and its affine description. Same chapter, flat.tex, Example labelled example-one-generator (source line 11474), invoked in the statement and at proof line 12250. Supplies A, f and J and the affine description of X, X', Z, E with which the whole proof works.
- Finite projective modules over a fibre product of rings. More on Algebra, Lemma labelled more-algebra-lemma-finitely-presented-module-over-fibre-product (Stacks tag 0D2J), invoked at proof lines 12286-12288. Supplies the finite projective module P' over A/(J cap fA) realising the given pair (P_1, P_2) with its compatibility; the Frobenius-twist construction of P from P' is inside the counted span.
- The fibre product of the colimits of the structure sheaves in the almost blowup square. Same chapter, Lemma labelled lemma-h-sheaf-colim-F (source line 11855), invoked at proof line 12263. Supplies the case P = A of the colimit identity, which the reduction of the proof reduces full faithfulness to.

Source reference. Original definition 0EVG: sources/stacks-flat/pinned-original-source.tex, source lines 11474–11474. Same-chapter pointer into the author environment example at source line 11474; invoked at proof line 12250 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EVK: sources/stacks-flat/pinned-original-source.tex, source lines 11855–11855. Same-chapter pointer into the author environment lemma at source line 11855; invoked at proof line 12263 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 0D2J: sources/stacks-chow/tags.txt, source lines 16384–16384. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12287 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let p be a prime number. Consider an almost blowup square X, X', Z, E in characteristic p as in Example 0EVG. Then the functor*

$$\operatorname{colim}_F \mathit{Vect}(X) \longrightarrow \operatorname{colim}_F \mathit{Vect}(Z) \times_{\operatorname{colim}_F \mathit{Vect}(E)} \operatorname{colim}_F \mathit{Vect}(X')$$

is an equivalence.

Proof. Let A, f, J be as in Example 0EVG. Since all our schemes are affine and since we have internal Hom in the category of vector bundles, the fully faithfulness of the functor follows if we can show that

$$\operatorname{colim} P \otimes_{A,F^N} A = \operatorname{colim} P \otimes_{A,F^N} A/J \times_{\operatorname{colim} P \otimes_{A,F^N} A/fA+J} \operatorname{colim} P \otimes_{A,F^N} A/fA$$

for a finite projective A-module P. After writing P as a summand of a finite free module, this follows from the case where P is finite free. This case immediately reduces to the case $P = A$. The case $P = A$ follows from Lemma 0EVK (in fact we proved this case directly in the proof of this lemma).

Essential surjectivity. Here we obtain the following algebra problem. Suppose P_1 is a finite projective A/J-module, P_2 is a finite projective A/fA-module, and

$$\varphi : P_1 \otimes_{A/J} A/fA + J \longrightarrow P_2 \otimes_{A/fA} A/fA + J$$

is an isomorphism. Goal: show that there exists an N, a finite projective A-module P, an isomorphism $\varphi_1 : P \otimes_A A/J \to P_1 \otimes_{A/J,F^N} A/J$, and an isomorphism $\varphi_2 : P \otimes_A A/fA \to P_2 \otimes_{A/fA,F^N} A/fA$ compatible with φ in an obvious manner. This can be seen as follows. First, observe that

$$A/(J \cap fA) = A/J \times_{A/fA+J} A/fA$$

Hence by More on Algebra, Lemma 0D2J there is a finite projective module P' over $A/(J \cap fA)$ which comes with isomorphisms $\varphi_1' : P' \otimes_A A/J \to P_1$ and $\varphi_2 : P' \otimes_A A/fA \to P_2$ compatible with φ. Since J is a finitely generated ideal and f-power torsion we see that $J \cap fA$ is a nilpotent ideal. Hence for some N there is a factorization

$$A \xrightarrow{\alpha} A/(J \cap fA) \xrightarrow{\beta} A$$

of F^N. Setting $P = P' \otimes_{A/(J\cap fA),\beta} A$ we conclude. □